INCH-POUND

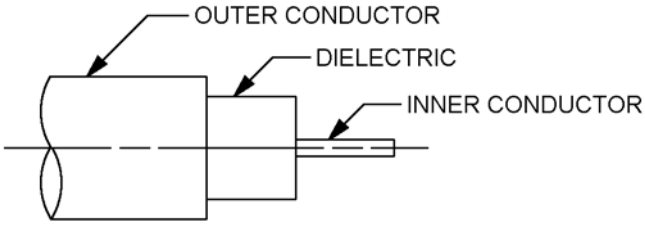
MIL-DTL-17/151A
w/AMENDMENT 1
20 April 2016
SUPERSEDING
MIL-DTL-17/151A
26 July 2006

DETAIL SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, COAXIAL, 0.047 INCH DIAMETER, SEMIRIGID, 50 OHMS, M17/151-00001 and M17/151-00002

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.



Inches	mm
.0005	0.013
.001	0.03
.002	0.05
.0113	0.287
.037	0.94
.047	1.19

* Metric conversions for table I

FIGURE 1. Configuration.

TABLE I. Description.

Part or Identifying Number (PIN)	Inner conductor	Dielectric core	Outer Conductor 1/	Weight, pounds per 1,000 feet (maximum)
M17/151-00001	Solid, silver-coated, copper clad steel .0113 inch ±.0005	Type F-1 diameter .037 inch ±.001	Seamless copper tubing, diameter .047 ± .001 inch	4.5
M17/151-00002	Same as above	Same as above	Seamless copper tubing tin plated in accordance with ASTM B545, .0003 inch thick, minimum. diameter .047 inch +.002, -.001 2/	4.8

1/ Use of recrystalized welded outer conductor is optional.
2/ The outer diameter is after plating.

AMSC N/A

FSC 6145

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Engineering information:

Continuous working voltage: 1,000 V rms, maximum.
Operating frequency: 20 GHz, maximum.
Velocity of propagation: 69.5 percent, nominal.
Power rating: See figure 2.
Operating temperature range: -40° to +100°C.

Inner conductor properties:
DC resistance (maximum at 20°C): 20.9 ohms per 100 feet.
Elongation: 10 percent minimum for class 40 A wire.
1.0 percent minimum for class 40 HS wire.
Tensile strength: 50 $klb_f/inch^2$ minimum for class 40 A wire.
110 $klb_f/inch^2$ minimum for class 40 HS wire.

Engineering notes: This cable is useful in critical RF performance applications (see connector series SMA, SMB and TNC in accordance with MIL-PRF-39012). This cable is generally manufactured in 20 foot lengths. Different lengths are available.

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:
Eccentricity: 8.5 percent, maximum.
Adhesion of conductors:
Inner conductor to core: 2 pounds minimum, 10 pounds maximum.
Outer conductor to core: 2 pounds, minimum.
Aging stability: Not applicable.
Stress crack resistance: Not applicable.

Outer conductor integrity: (+175°C for recrystalized outer conductor cables) Four specimens, approximately 2 feet long, shall be cut from the sample unit. The specimens shall be suspended in a heat chamber without touching one another or the walls of the chamber and conditioned for 1 hour 30 minutes minimum at the specified maximum operating temperature ±5ºC. Heated air shall be circulated so as to maintain a uniform test temperature. After the conditioning period, the specimens shall be removed from the heat chamber and conditioned at room ambient temperature for 4 hours minimum. The specimen shall then have one end immersed into molten solder to a depth of 0.5 inch minimum for 15 seconds minimum. The molten solder temperature shall be +230°C minimum. After the conditioning period, the specimens shall be removed from the molten solder and conditioned at room ambient temperature for 1 hour minimum. Examine the specimens for cracks, flaws, or other damage in the outer conductor material.

Cold bend: Not applicable.
Dimensional stability: +125°C ±5°C.
Core to Conductor: .015 inch, maximum.

Contamination: Not applicable.
Bendability: Mandrel diameter: .25 inch, maximum.
Flammability: Not applicable.
Weight: See table I.

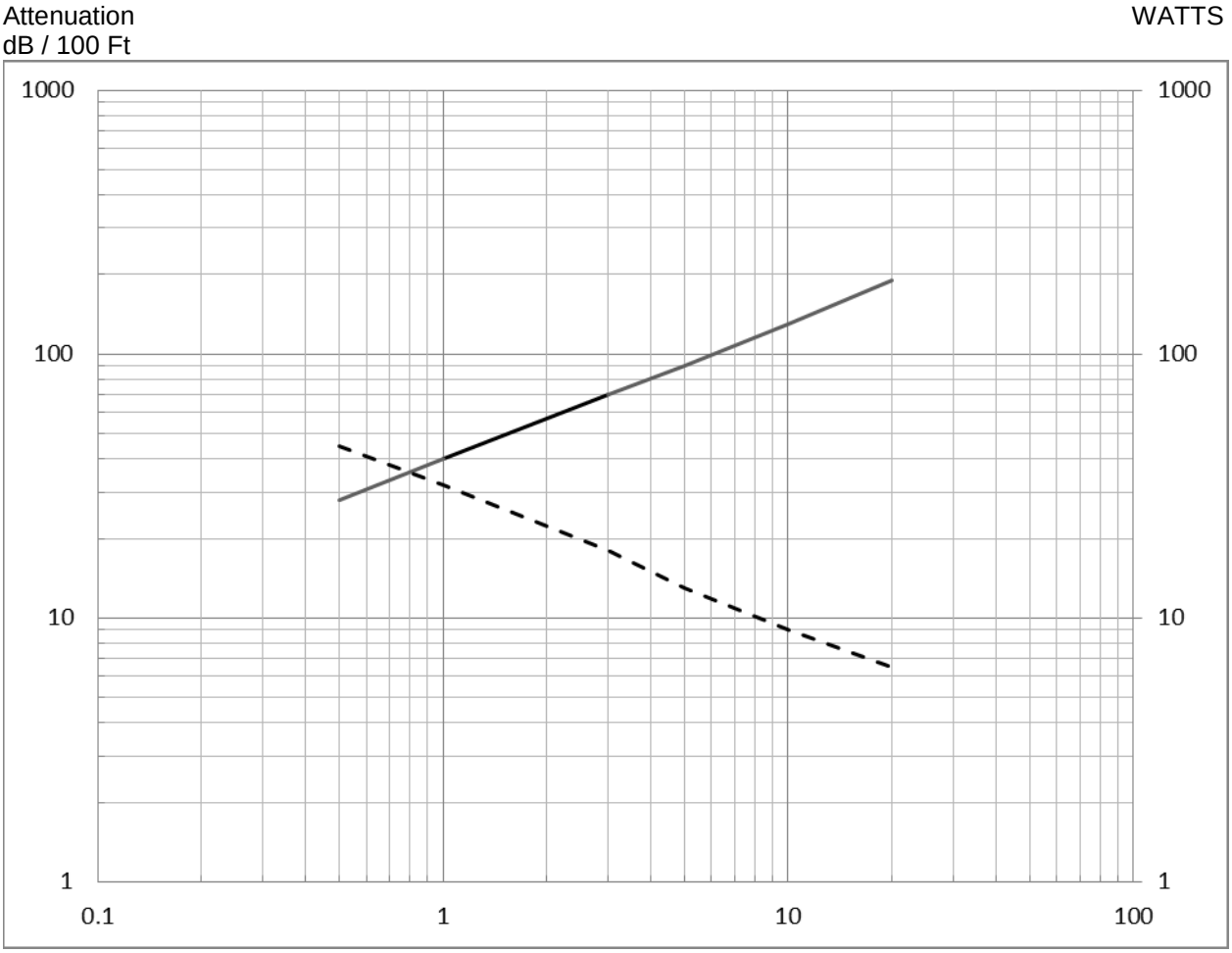
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Electrical:

Test frequency: 500 MHz to 20 GHz.
Spark test: Not applicable.
Voltage withstanding: 2,000 Vrms, minimum.
Insulation resistance: Not applicable.
Corona extinction voltage: 1,000 V rms, minimum.
Characteristic impedance: 50.0 ohms ± 2.5.
Attenuation: See figure 2.
Structural return loss: See figure 3.
Capacitance: 32 pF per foot, maximum.
Capacitance stability: Not applicable.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Phase stability: Not applicable.
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable.

PIN: See table I.

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Maximum attenuation ———————— (Test requirements shall be as noted as line indicates on graph)

Maximum power - - - - - - - - - - - - - -

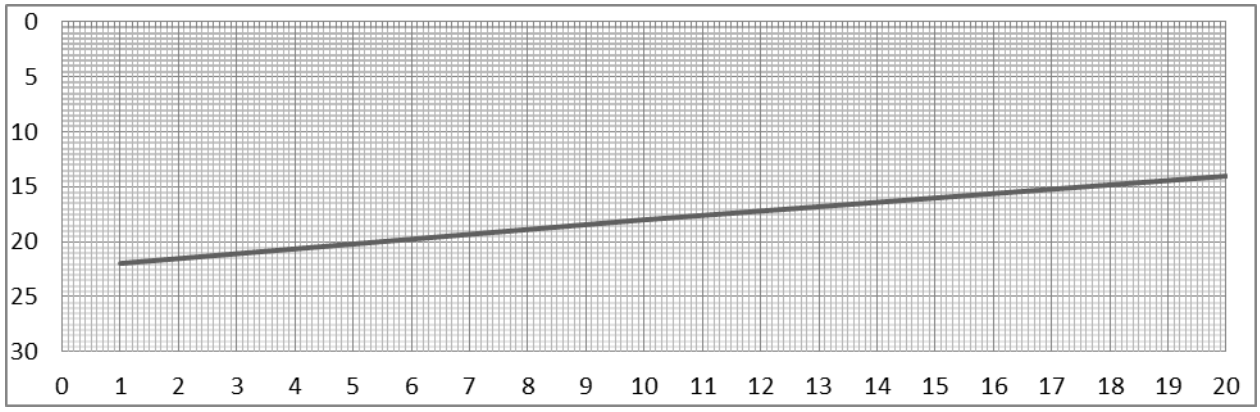
At 25°C sea level.

Attenuation		Power
MHz	dB	Watts
500	28	45
1000	40	32
3000	70	18
5000	90	13
10000	130	9
20000	190	6.5

FIGURE 2. Power rating and attenuation.

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MINIMIM STRUCTURAL RETURN LOSS



SWR	REFLECTION COEFFICIENT	RETURN LOSS dB	SWR	REFLECTION COEFFICIENT	RETURN LOSS dB
17.3910	.8913	1	1.3767	.1585	16
8.7242	.7943	2	1.3290	.1413	17
5.8480	.7079	3	1.2880	.1259	18
4.4194	.6310	4	1.2528	.1122	19
3.5698	.5623	5	1.2222	.1000	20
3.0095	.5012	6	1.1957	.0891	21
2.6146	.4467	7	1.1726	.0794	22
2.3229	.3981	8	1.1524	.0708	23
2.0999	.3548	9	1.1347	.0631	24
1.9250	.3162	10	1.1192	.0562	25
1.7849	.2818	11	1.1055	.0501	26
1.6709	.2512	12	1.0935	.0447	27
1.5769	.2239	13	1.0829	.0398	28
1.4985	.1995	14	1.0736	.0355	29
1.4326	.1778	15	1.0653	.0316	30

Structural	Return loss
MHz	dB
1000	22
10000	18
20000	14

FIGURE 3. Structural return loss.

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Referenced documents. In addition to MIL-DTL-17, this document references the following:

ASTM B545
MIL-PRF-39012

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2016-003)

Review activities:
Army - AR, AT, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99
DLA - IS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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